

IoT-Brain: Grounding LLMs for Semantic-Spatial Sensor Scheduling

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ABSTRACT

Intelligent systems powered by large-scale sensor networks are shifting from predefined monitoring to intent-driven operation, revealing a critical *Semantic-to-Physical Mapping Gap*. While large language models (LLMs) excel at semantic understanding, existing perception-centric pipelines operate retrospectively, overlooking the fundamental decision of what to sense and when. We formalize this proactive decision as *Semantic-Spatial Sensor Scheduling (S^3)* and demonstrate that direct LLM planning is unreliable due to inherent gaps in representation, reasoning, and optimization. To bridge these gaps, we introduce the *Spatial Trajectory Graph (STG)*, a neurosymbolic paradigm governed by a "verify-before-commit" discipline that transforms open-ended planning into a verifiable graph optimization problem. Based on *STG*, we implement *IoT-Brain*, a concrete system embodiment, and construct *TopoSense-Bench*, a campus-scale benchmark with 5,250 natural-language queries across 2,510 cameras¹. Evaluations show *IoT-Brain* boosts task success rate by 37.6% over the strongest search-intensive methods while running nearly 2× faster and using 6.6× fewer prompt tokens. In real-world deployment, it approaches the reliability upper bound set by reducing 4.1× network bandwidth, providing a foundational framework for LLMs to interact with the physical world with unprecedented reliability and efficiency.

CCS CONCEPTS

- **Computing methodologies** → **Artificial intelligence**;
- **Computer systems organization** → **Sensor networks**.

KEYWORDS

Large Language Models, IoT Networks, Sensor Scheduling

1 INTRODUCTION

The proliferation of large-scale sensor networks in smart cities and industries is catalyzing a paradigm shift towards intelligent automation[42, 49, 55]. This leap from traditional coverage-driven optimization[11, 32, 58, 75] to real-time semantic goal satisfaction introduces a profound, yet under-explored challenge. A simple request like "*Can you help me check my wallet between the library and the gym?*" exposes a stark divide between the vagueness of human language and the precise physical operations a sensor network must

execute. We term this the *Semantic-to-Physical Mapping Gap* (Fig. 1(a)), a fundamental hurdle that renders conventional task-specific models and rigid scripts ineffective.

The advent of large language models (LLMs) has provided a powerful engine for semantic interpretation [19, 26, 50, 52], offering a promising path forward. Recent work on "Penetrative AI" further shows that LLMs can reason directly over raw sensor data [18, 59, 76, 81]. Yet these advances largely remain within operates retrospectively, assuming the relevant sensor streams are already available, what we term *Reactive Perception* (Fig.1(c)). This perspective overlooks a more fundamental upstream challenge, *Proactive Scheduling*, that is paramount in large-scale deployments. Before any meaningful perception can occur, LLMs must decide what to sense and when to attend. The decision precedes and enables all subsequent perception but has received limited systematic study [5, 35, 43]. We formalize this pivotal challenge as the *Semantic-Spatial Sensor Scheduling (S^3)* problem.

Solving the S^3 problem with off-the-shelf LLMs is far from straightforward. Our preliminary study (§2.2) tasks an LLM with end-to-end scheduling in a real-world topological environment, revealing three fundamental challenges:

- (1) **Symbol-to-Semantic Chasm**. LLMs' native shortcoming in comprehending raw, machine-oriented symbolic topologies prevents them from building an effective world model, slashing their planning success by over 5× compared to when provided with structured, human-readable knowledge.
- (2) **Inferential Leap from Points to Paths**. LLMs' profound difficulty in inferring topological relationships like connectivity from disconnected symbols causes even a perfectly informed model to achieve merely 26% trajectory coverage, leading to fragmented and unsound paths.
- (3) **Optimization Shortfall in LLM Planning**. The inherent "satisficing" nature of LLMs leads to resource-heavy plans, exhibiting up to 45% redundant sensor overlap even with structured guidance and compounding to a staggering 48× token overhead when processing raw symbolic data.

Core idea: verify-before-commit. This work aims to bridge the gap between an LLM's high-level semantic reasoning and the need for resource-efficient, physically grounded sensor scheduling. Our insight stems from observing LLMs' native behavior on the S^3 problem, showing that their plans are often ungrounded and glaringly misaligned with sensor-network constraints. By contrast, when supplied with pre-vetted, task-relevant topological knowledge, LLM-derived

¹Project page: <https://anonymous.4open.science/r/IoT-Brain-E465>

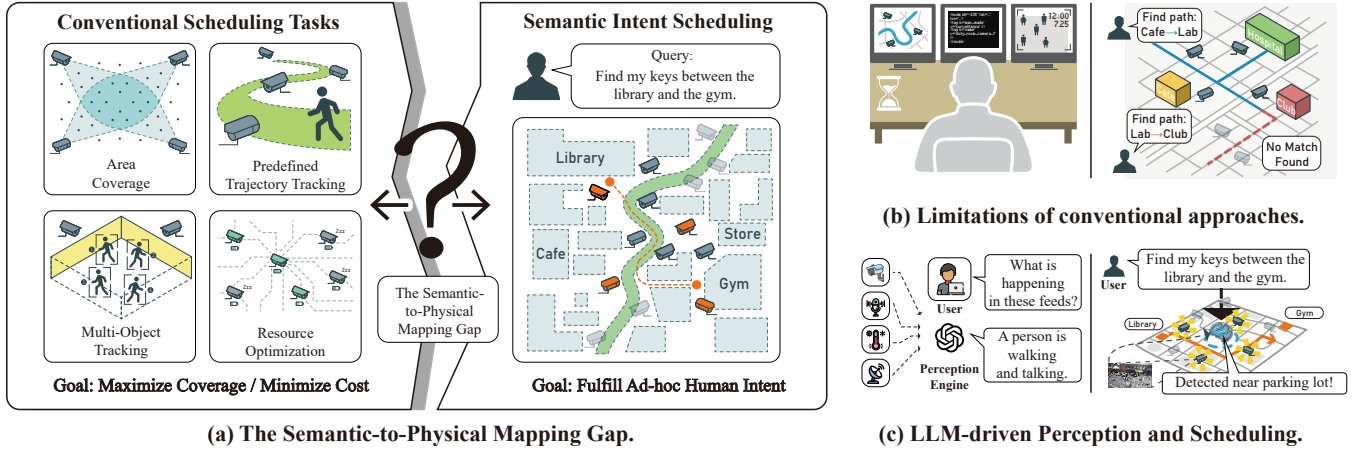


Figure 1: The challenge of the S^3 problem.

plans become both reliable and efficient. Such oracle-like access, however, is infeasible in dynamic deployments. In resource-sensitive environments, executing a speculative, unverified plan is prohibitively expensive. We therefore adopt a paradigm in which the LLM must proactively and autonomously discover and validate the necessary grounding knowledge through direct interaction with the physical world before any operational decision is taken. We term this disciplined approach "verify-before-commit", a principle requiring that all semantic hypotheses be fully validated against reality before they are deemed executable.

STG design. The "verify-before-commit" principle's progressive translation of intent into action inherently embodies a search for an optimal topological pathway within semantic constraints. This complex search process mirrors the paradigm of graph construction[15, 70, 78], which we formalize as the *Spatial Trajectory Graph (STG)*, a neurosymbolic paradigm that operates through a multi-stage workflow. The paradigm first mandates the structuring of ambiguous intent into a verifiable, hypothesized graph. It then requires grounding of the graph through an iterative validation loop against a physical world model, methodically transforming uncertainty into verified facts. Finally, the paradigm concludes with the optimization of the now-verified graph into a resource-aware, dynamic execution plan. This principled decomposition systematically separates semantic interpretation from deterministic validation and scheduling.

IoT-Brain Implementation. We implement the STG paradigm in *IoT-Brain*, a system engineered for robust, real-world interaction. Instead of a monolithic pipeline, *IoT-Brain* adopts a modular architecture that leverages an LLM's advanced tool-calling and programming capabilities[34, 57]. It employs a reactive, tool-based loop to continuously ground its semantic reasoning against our physical world model, effectively transforming abstract hypotheses into verifiable facts. This design strategically decouples the LLM's high-level semantic inference from the deterministic, resource-conscious tasks

of scheduling and sensor control. The entire workflow is further accelerated by shared memory modules that cache executable reasoning evidence, amortizing the cost of interaction across sessions. To evaluate *IoT-Brain*, we also constructed *TopoSense-Bench*, a large-scale benchmark tailored to the S^3 problem featuring a university-scale digital twin with 2,510 cameras and 5,250 real-world queries.

Contributions of our work are summarized as follows:

- We identify and formalize the S^3 problem, a critical yet overlooked challenge in intent-driven sensor networks, and introduce the *Spatial Trajectory Graph (STG)*, a neurosymbolic paradigm that grounds LLM reasoning in a verifiable, physical-world structure.
- We design and implement *IoT-Brain*, our concrete system instantiation of STG, and we construct and release *TopoSense-Bench*, a large-scale benchmark with 5,250 real-world queries to catalyze future research in this domain.
- We conduct evaluations of *IoT-Brain* on our *TopoSense-Bench* and a physical testbed. Experimental results demonstrate the superiority of STG. On the benchmark's most complex tasks, *IoT-Brain* boosts task success by 37.6% over the strongest search-intensive methods[14, 56] while running nearly $2\times$ faster and using $6.6\times$ fewer prompt tokens. Furthermore, in live deployments, our system approaches the reliability of a resource-agnostic upper-bound approach while consuming $4.1\times$ less network bandwidth.

2 BACKGROUND & MOTIVATION

We first survey the sensor-scheduling landscape, contrasting coverage-driven methods with emerging LLMs' capabilities to expose a critical gap (§2.1). Then we present a preliminary study that decomposes LLM-based proactive scheduling into three fundamental gaps (§2.2). Next, we motivate a new planning paradigm that addresses these gaps (§2.3).

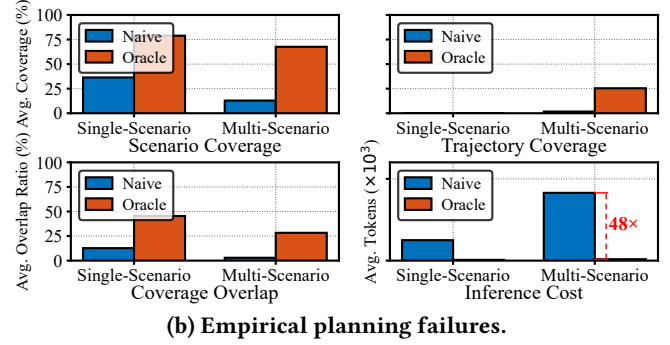
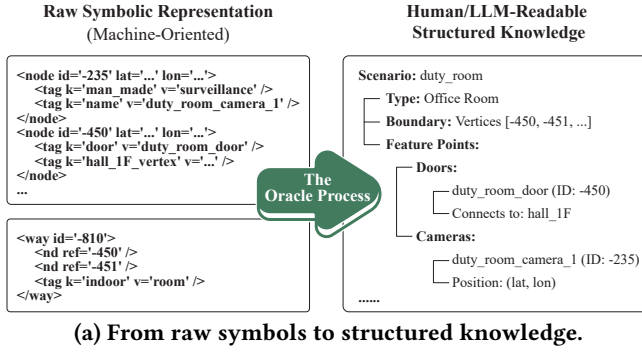


Figure 2: The representation gap and resulting planning failures.

2.1 Background

Classical research in sensor networks has centered on coverage-driven tasks, such as maximizing sensor coverage or tracking objects within a pre-defined field-of-view [11, 13, 38]. While effective for well-defined objectives, this paradigm is ill-suited for intent-driven queries specified in natural language. In practice, such ad-hoc semantic tasks are relegated to rudimentary solutions, typically laborious human-in-the-loop operations [28, 47, 54] or brittle fixed-topology scripts [10, 30]. As illustrated in Fig. 1(b), both approaches break down in large-scale, dynamic settings [64, 67, 74]. Manual operation does not scale [20], and hard-coded scripts fail to generalize to the fluid nature of human intent [51, 69].

The semantic understanding of LLMs offers a promising route beyond these limits. Pioneering systems show LLMs can parse complex sensor data and interact with the physical world, opening a new frontier for AIoT [18, 23, 59, 62, 81]. Yet, the current LLM-for-AIoT landscape is imbalanced. As shown in Fig. 1(c), most work concentrates on *Reactive Perception*, where models analyze pre-collected sensor streams. We instead tackle *Proactive Scheduling*, the upstream problem of deciding which sensors to activate and when. This shift from passive interpretation to active, goal-directed participation is essential for truly autonomous AIoT systems, and the transition is nontrivial, exposing fundamental challenges in grounding LLM reasoning in the physical world.

2.2 Preliminary Study

To dissect why the leap from perception to scheduling is challenging for LLMs, we conducted a preliminary study designed to answer a core question. *Can LLMs effectively plan routes and schedule sensors when given raw, symbolic topological data?* To investigate this, we constructed a dedicated testbed using real-world topological data from OpenStreetMap (OSM) [27], encompassing five multi-scenario buildings and a set of 373 manually curated queries that require both spatial understanding and path planning.

We compared two approaches. In the Naive setting, the LLM is prompted directly with machine-oriented OSM text

(e.g., XML-like nodes and ways), and is asked to produce a sensor activation plan. In the Oracle setting, which serves as an upper bound, the same symbolic data is preprocessed into a structured, human-readable knowledge base that makes locations, connections, and salient features explicit (see Fig. 2(a)) for which the LLM then uses for reasoning. We assessed each approach on its ability to produce correct and efficient schedules, measuring scenario coverage, trajectory coverage, resource overlap, and token consumption. The results, summarized in Fig. 2(b), reveal three tightly coupled gaps that together impede reliable LLM-based scheduling.

Gap 1: Symbol-to-Semantic Chasm. The disparity between the Naive and Oracle settings reveals a representation mismatch. The impact of this chasm is stark. When the LLM is provided with structured, human-readable knowledge, scenario coverage boosts by over 2.1× on single-scenario tasks and a remarkable 5× on multi-scenario tasks compared to the Naive approach. Because LLMs are trained on natural language rather than machine-oriented symbolic topologies, they struggle to parse OSM-style structures [21, 22, 60] and therefore cannot assemble a usable world model. Planning performance consequently collapses, underscoring the need for a dedicated symbol-to-semantic translation layer.

Gap 2: Inferential Leap from Points to Paths. Trajectory coverage lays bare a deeper reasoning failure. In the Naive setting, it is essentially zero. Even with an explicit, intelligible map, the Oracle reaches only 26% on multi-scenario tasks. This shortfall reflects a deficit in multi-step path formation across large topologies. LLMs can reference named places and operate over simple pre-specified graphs, yet they rarely assemble the connectivity constraints that turn local doorways into a valid end-to-end route. Rather than simply providing structured data and expecting a correct plan, a practical system must actively scaffold reasoning by constructing connectivity, verifying reachability, and validating long-horizon trajectories within the spatial graph.

Gap 3: Optimization Shortfall in LLM Planning. Beyond representation and reasoning, planning quality falters at the optimization level. Even with perfect topological semantics, the Oracle still yields inefficient schedules, with overlap

reaching 45%. Without structured guidance, the inefficiency compounds dramatically. On multi-scenario tasks, the Naive setting expends a staggering 48× more tokens than the Oracle. These patterns clearly indicate LLMs inherently tend to satisfy rather than optimize, producing plausible yet resource-heavy plans [44, 73]. Therefore, a practical system should capitalize on LLM's semantic strengths while delegating global resource efficiency to deterministic algorithms[6].

2.3 Motivation & Core Ideas

The preceding findings yield a crucial insight that a reliable and efficient semantic scheduling system cannot treat the LLM as an end-to-end black-box planner. These inherent gaps in representation, reasoning, and optimization necessitate a paradigm to explicitly structure the LLM's role.

This motivates our core idea to replace monolithic opaque planning with a structured and verifiable workflow centered on the *Spatial Trajectory Graph (STG)*. Our neurosymbolic STG paradigm decomposes the intractable scheduling problem into a principled three-stage process, each instantiated by a corresponding phase in our *IoT-Brain* system. 1) *Intent Formalization* (§4.2) first leverages an LLM's semantic competence to translate a user's ambiguous query into a hypothesized STG, a structured but unverified blueprint of intent. 2) *Feasibility Grounding* (§4.3) then enters an iterative "verify-before-commit" loop where each element of the blueprint is systematically validated against the physical world model until a fully consistent and grounded STG is produced. 3) *Optimal Synthesis* (§4.4) finally compiles the now-verified blueprint into a resource-optimal sensor activation plan and executes it adaptively using a perception-in-the-loop mechanism to handle unfolding real-world dynamics. The multi-stage process provides a systemically verifiable solution for LLM-based grounding, making a significant step towards enabling truly intelligent AIoT systems.

3 S³ AND STG FORMULATION

We first formally define the *Semantic-Spatial Sensor Scheduling (S³)* problem as a principled mapping from high-level user intent to a resource-aware dynamic activation plan (§3.1). Building on this formulation, we introduce the *Spatial Trajectory Graph (STG)*, a paradigm that grounds free-form language into an optimization-ready spatial graph to systematically address this fundamental challenge (§3.2).

3.1 The S³ Problem

Environment and Query. We consider a large, densely instrumented site where a user issues a natural language query Q_{NL} . The environment is represented by a comprehensive world model $W = (\mathcal{G}_{spatial}, \mathcal{G}_{sensor}, \Phi)$ where $\mathcal{G}_{spatial}$ is a labeled spatial graph of locations and traversability, $\mathcal{G}_{sensor}$ is

a device graph of sensors and their capabilities, and Φ links the two by encoding sensor visibility and geometry.

Semantic Compilation. Given Q_{NL} and W , semantic compilation yields a set of verifiable spatiotemporal predicates $\Omega(Q_{NL}, W)$ specifying anchors, regions, relational constraints, and temporal bounds (e.g., "11:30 a.m."). Let $\Pi(\Omega)$ denote the spatiotemporal witnesses, the trajectories that must be observed within a specific timeframe to satisfy the query.

Plans and Objective. Answering the query requires generating a dynamic activation plan $P(t)$, a time-varying set of sensors intended to reconstruct a witness trajectory $\tau \in \Pi(\Omega)$. A plan is feasible if its collective observation over time, denoted as $\mathcal{P} = \bigcup_t P(t)$, completely covers at least one witness τ . The objective is to find an optimal plan $P^*(t)$ that maximizes a fidelity-cost trade-off, formalized as:

$$P^*(t) = \arg \max_{P(t) \subseteq \mathcal{G}_{sensor}} \left(\mathcal{F}(\mathcal{P}; \Pi(\Omega), \Phi) - \lambda \int \text{Cost}(P(t)) dt \right), \quad (1)$$

where fidelity $\mathcal{F}$ measures the quality of the reconstructed trajectory, monotonically increasing with its completeness and accuracy. The $\text{Cost}(P(t))$ function captures all resource expenditures, including the number of active sensors, activation duration, and redundant spatial overlap.

Inherent Requirements. While Eq. 1 specifies the optimization objective, solving it directly with LLMs is intractable given the model's intrinsic limits. Informed by our preliminary study (§2.2), a viable LLM-based approach must satisfy three fundamental requirements to align model capability with problem demands. 1) *Semantic Grounding*. The solution must anchor the LLM's linguistic outputs to the physical world. Fuzzy descriptions like "near the main entrance" must be unambiguously resolved to concrete entities in W to become actionable. 2) *Topological Feasibility*. The solution must enforce topological validity on the LLM's generated plans. Any long-horizon trajectory τ must be explicitly verified as physically traversable within $\mathcal{G}_{spatial}$, precluding the LLM from hallucinating impossible paths. 3) *Resource Efficiency*. The solution must decouple planning from optimization. Given that LLMs are satisficers, not optimizers, a practical system must delegate the selection of a resource-efficient schedule to deterministic algorithms. These requirements motivate a paradigm that structures and constrains the LLM's role, moving beyond brittle end-to-end planning.

3.2 The STG Paradigm

End-to-end sensor selection couples semantics, topology, resources, and timing, hindering verification of intermediate assumptions. To make decisions verifiable and cost-aware, we introduce STG. STG decouples semantic interpretation, topological grounding, and resource optimization by inserting an explicit spatiotemporal trajectory graph where candidate hypotheses are checked before commitment. This blueprint

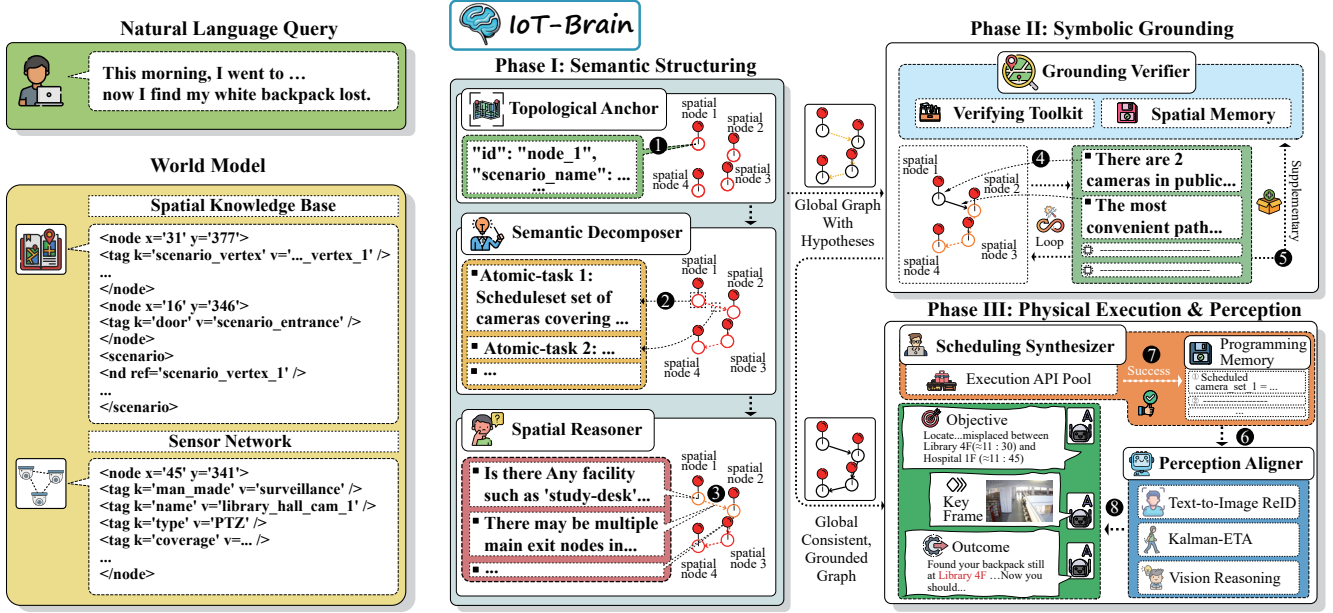


Figure 3: The system overview and workflow of IoT-Brain.

transforms a user's intent into a structured trajectory, the trajectory into verified spatiotemporal facts, and these facts into an optimized, dynamic activation schedule.

STG Definition. An *STG* is a quadruple $G = (V, E, \tau_V, \sigma_t)$ that provides a verifiable specification for a dynamic plan. It comprises: (i) a connected subgraph (V, E) of relevant locations; (ii) a verified spatial witness path $\tau_V = (v_1, \dots, v_m)$ that satisfies the spatial predicates; and (iii) a dynamic sensor scheduling function σ_t , which maps each spatial node $v_i \in \tau_V$ to a set of sensors to be activated at a dynamically determined time t_i . The induced dynamic plan is thus $P(t) = \sigma_t(v_i)$ for $t = t_i$. An *STG* is hypothesized (G_0) when its spatial path $\tau_{V,0}$ is unverified, and becomes grounded ($G_\star$) only after τ_V is validated against the world model W .

Objective Reparameterization. The *STG* structure recasts the original optimization over dynamic plans $P(t)$ as a more tractable, two-stage process. First, it searches over the space of grounded spatial paths ($G_\star$) to find an optimal τ_V^* . Second, it determines the optimal temporal scheduling (σ_t^*) along that path. The objective is formalized as:

$$(\tau_V^*, \sigma_t^*) = \arg \max_{(\tau_V, \sigma_t)} \left(\mathcal{F}(\tau_V, \sigma_t) - \lambda \text{Cost}(\tau_V, \sigma_t) \right), \quad (2)$$

where the optimal dynamic plan $P^*(t)$ is derived from (τ_V^*, σ_t^*) . This reframing is pivotal because it separates the verifiable, static spatial planning from the adaptive, online temporal scheduling, making the problem tractable.

Principled Inference Workflow. The reframing supports a three-stage workflow guided by the "verify-before-commit" principle. i) *Intent Formalization.* From the user query, construct an ungrounded graph G_0 populated with candidate locations and a hypothesized spatial path $\tau_{V,0}$. ii) *Feasibility*

Grounding. Enter an iterative loop that validates spatial hypotheses in $\tau_{V,0}$ against the world model W , disambiguates semantics, and enforces topological feasibility until a fully grounded spatial path $G_\star$ emerges. iii) *Optimal Synthesis.* On the grounded path $G_\star$, compute the optimal dynamic scheduling function σ_t^* maximizes the objective in Eq. 2. This yields a verifiably resource-aware dynamic activation plan $P^*(t)$. Within this workflow, the LLM proposes and refines spatial hypotheses, while the correctness of the spatial path and the optimality of the spatiotemporal schedule are secured by explicit checks and deterministic solvers.

4 IOT-BRAIN SYSTEM ARCHITECTURE

To instantiate the *STG* paradigm, we implemented *IoT-Brain*, a modular framework that turns high-level semantic queries into verifiable, resource-aware sensor activation plans. As depicted in Fig. 3, *IoT-Brain* follows a three-stage pipeline comprising Semantic Structuring (§4.2), Symbolic Grounding (§4.3), and Adaptive Execution and Perception (§4.4).

4.1 System Workflow

IoT-Brain takes two inputs, a natural language query Q_{NL} expressing the user's ad-hoc intent and a world model W that combines a spatial knowledge base with a sensor-network map, and then processes them through a three-stage pipeline. **Semantic Structuring.** The workflow begins with the *Topological Anchor* ①, which converts the free-form query into a structured preliminary blueprint by binding salient places in Q_{NL} to candidate nodes and links, thereby seeding an initial graph scaffold. Building on the skeleton, the *Semantic Decomposer* ② factors the user's goal into a sequence of atomic

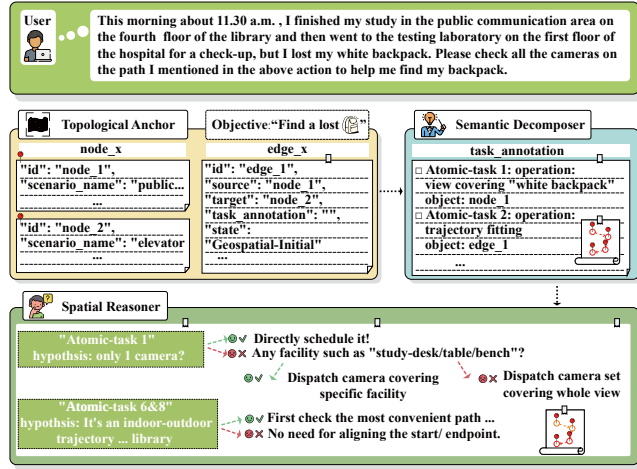


Figure 4: Dataflow of the Semantic Structuring phase. tasks such as covering key zones or tracing admissible paths. The *Spatial Reasoner* ③ then augments the plan with concrete, verifiable hypotheses about entrances, corridors, and visibility relations. The outcome is a hypothesized global graph that is comprehensive yet not verified.

Symbolic Grounding. An iterative "verify-before-commit" loop establishes reliability. The *Grounding Verifier* ④, aided by a *Verifying Toolkit*, checks each hypothesis against the world model, retaining confirmed items while refining contradictions until a consistent grounded graph emerges. To accelerate subsequent queries, the *Spatial Memory* ⑤ module caches stable facts discovered during grounding.

Adaptive Execution and Perception. Once the blueprint is grounded, the *Scheduling Synthesizer* ⑥ compiles it into an executable script. This script invokes heuristic solvers from our *Execution API Pool* to generate a resource-optimal schedule. Successful query-script pairs are cached in *Programming Memory* ⑦ for reuse via in-context learning[45, 83]. The program then activates the *Perception Aligner* ⑧ to execute the schedule adaptively. The *Aligner* uses text-guided re-identification for target initialization, a Kalman-ETA[3] module for just-in-time sensor activation, and VLM-based reasoning to exit early once the query is resolved.

User scenario. Consider a user who has misplaced his backpack. After a brief, natural language query describing the situation and path, *IoT-Brain* processes the request autonomously. The system intelligently constructs a verified, topologically sound search strategy, compiles it into an executable script, and finally orchestrates the required cameras in real time to trace the route, locate the item, and report its final position.

4.2 Intent-to-Blueprint Structuring

The initial phase of *IoT-Brain* converts the unstructured user query Q_{NL} into a structured representation of intent and context. This process, termed *Semantic Structuring*, progressively builds a hypothesized STG, which is a machine-interpretable

blueprint derived from the query but not yet verified against the world model. Fig. 4 depicts the three-agent pipeline that orchestrates the language-to-graph transformation.

① **Topological Anchor.** The pipeline begins with the *Topological Anchor*, a coarse-grained parser that identifies key geographical entities and their high-level relationships within the query. For a request like "I walked from the library to the gym", the *Anchor* extracts "library" and "gym" as primary spatial anchors. These textual mentions instantiate a set of candidate vertices and edges by querying the world model's Spatial Topology ($\mathcal{G}_{spatial}$). The agent focuses only on explicitly mentioned entities to form a high-level, sparse graph structure. Its output formally constitutes the initial, unverified vertices V_0 and edges E_0 of a nascent STG, representing the initial hypothesis of the user's spatial context.

② **Semantic Decomposer.** While the *Anchor* provides a spatial skeleton, it lacks a concrete plan of action. The *Semantic Decomposer* serves as a fine-grained task planner that builds upon this skeleton. It interprets the user's ultimate goal (e.g., "find my lost keys") and decomposes it into a detailed, logical sequence of atomic sub-tasks. This process effectively defines the spatial witness walk, $\tau_{V,0} = (v_1, \dots, v_m)$, a hypothesized trajectory through the graph that, if followed, would satisfy the user's intent. A typical decomposition confirms the starting area, traverses the route to the destination, and searches the destination zone. The resulting graph is enriched with $\tau_{V,0}$ and associated task annotations.

③ **Spatial Reasoner.** Although annotated with a witness walk, the graph still rests on a series of implicit assumptions. The final agent in this phase, the *Spatial Reasoner*, makes these assumptions explicit by populating the graph with testable hypotheses. For each step v_i in the witness walk $\tau_{V,0}$, the *Reasoner* analyzes the underlying topological logic to generate critical questions for the plan's feasibility. It might hypothesize, for example, that a specific, unmentioned exit door must be used for a transition, or that certain facilities within a room could constrain the search.

Each hypothesis is cast as a precise, verifiable proposition about the world model (e.g., "Verify existence of 'main_door' in 'library'"), with explicit scope, expected relations, and admissible evidence sources. This step leaves the sensor scheduling intentionally unspecified. The three-agent pipeline then returns a comprehensive hypothesized spatial path, encapsulated in an initial STG, $G_0 = (V_0, E_0, \tau_{V,0}, \emptyset)$. The resulting spatial blueprint is aligned with the user's semantic intent and fully annotated with verifiable assumptions, preparing it for the *Symbolic Grounding* phase.

4.3 Hypothesis-to-Fact Grounding

Even a detailed hypothesized graph remains non-actionable because its elements are unverified. The *Symbolic Grounding*

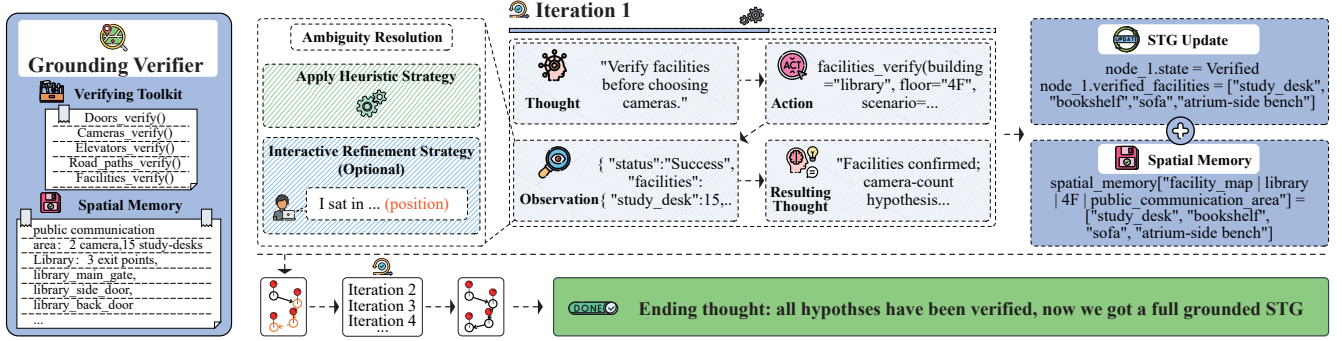


Figure 5: Workflow of the Hypothesis-Verification Loop.

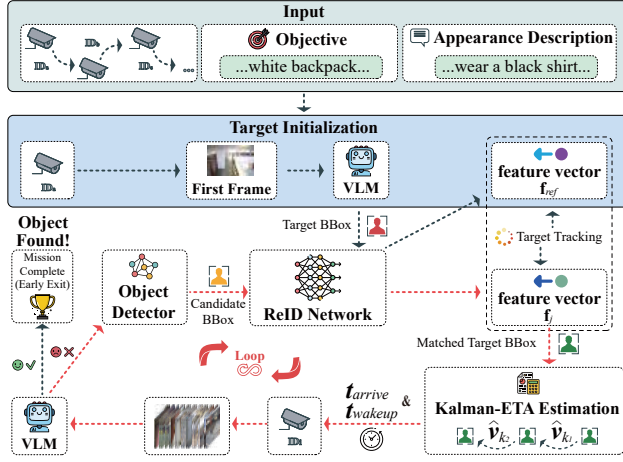


Figure 6: Workflow of the Perception Aligner.

phase converts the hypothesized STG , G_0 into a grounded STG , $G_\star$ by rigorously testing each hypothesis against the world model W . As shown in Fig. 5, a "verify-before-commit" loop promotes facts, prunes contradictions, and resolves ambiguities until the graph is consistent with W .

4 The Hypothesis-Verification Loop. Grounding hinges on a hypothesis-verification loop executed by the *Grounding Verifier*. The Verifier iterates through a Thought, Action, and Observation (TAO) cycle [65, 80] to systematically resolve uncertainties. In each iteration, it selects an unverified hypothesis, formulates a precise "thought" describing the test and success criteria, and performs an "action" by invoking the appropriate function from the *Verifying Toolkit* (e.g., `doors_verify()`) to query the spatial knowledge base. The resulting "observation", a hard fact from the world model, updates STG by either confirming the hypothesis or exposing ambiguities that require refinement.

Resolving Ambiguity. A critical challenge within this loop is handling ambiguity, as an observation from the *Verifying Toolkit* may not be definitive. For instance, a query for the "main entrance" may yield several valid doors. *IoT-Brain* addresses such cases with two complementary strategies. In the default heuristic strategy, the system proceeds by applying a goal-consistent heuristic, for instance, selecting the door

that lies on the shortest feasible path to the next waypoint while respecting visibility and access constraints. This choice preserves autonomy and mirrors human decision-making under uncertainty. As an optional enhancement, the interactive refinement strategy presents the alternatives to the user for quick disambiguation (e.g., "I found two main entrances. Which one did you use?"). Interactive refinement is useful in certain settings, but unless otherwise stated, our evaluation in Sec. 5 focuses on the robust autonomous strategy.

5 Amortized Verification via Caching. Throughout the iterative process, successfully verified facts are cached according to their associated locations in a *Spatial Memory* module, a technique inspired by classic indexing and memoization in knowledge-intensive systems [40]. This mechanism amortizes the cost of repeated queries for the same entities, accelerating future verification at the same site. The verification loop terminates once all hypotheses have been resolved. The final output is the grounded STG , $G_\star$, which represents a verified spatial blueprint. Within this blueprint, every node, edge, and the contained spatial path τ_V are verifiably consistent with the physical world, forming a reliable foundation for the subsequent scheduling optimization.

4.4 From Plan to Optimized Action

With a fully grounded spatial blueprint $G_\star$ in place, the final phase of *IoT-Brain* translates it into dynamic action and perception in the physical world. The workflow comprises two components to synergistically achieve resource optimization. The first is a *deterministic compiler* that synthesizes an executable and resource-aware plan. The second is an adaptive executor that manages real-time operation.

6 Scheduling Synthesizer. The *Scheduling Synthesizer* functions as a graph-to-code compiler, transforming the grounded $G_\star$ into an executable Python script. This stage operationalizes the resource optimization objective in Eq. 2 by invoking specialized functions from the *Execution API Pool* that encapsulate proven heuristics. For example, to cover a trajectory segment, it calls `indoor_path_camera_search()`,

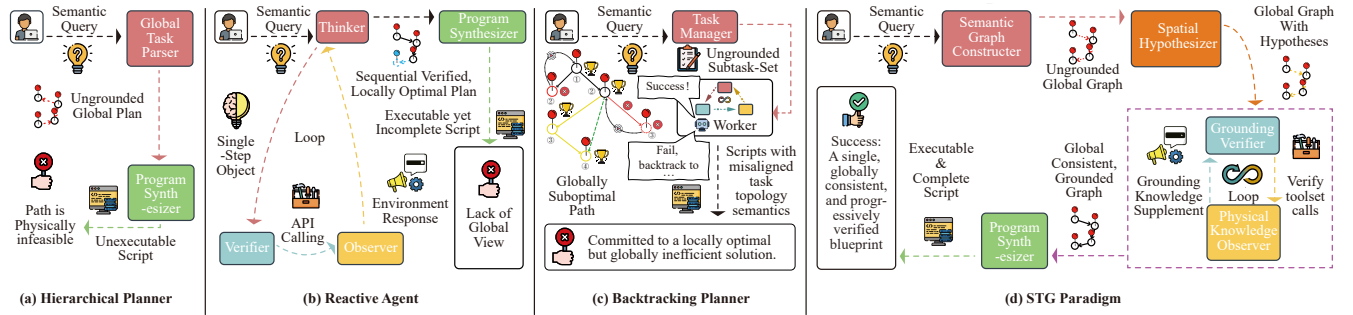


Figure 7: Conceptual workflow comparison of agentic paradigms. We contrast the brittle Hierarchical approach, the myopic Reactive agent, and the costly Backtracking planner with our structuring verifiable STG paradigm.

which solves a set cover style problem to return a minimal sensor set[86]. By embedding optimization within a deterministic API layer, the compiler produces schedules that honor topological constraints and visibility assumptions while remaining resource efficient. Successful query-script pairs are cached in *Programming Memory* ⑦ to accelerate future compilations on semantically-related tasks.

⑧ **Perception Aligner.** A static script is insufficient for dynamic, real-world execution. The Perception Aligner therefore operates as an online executor, tightly coupling perception with control to keep only necessary sensors active. As depicted in Fig. 6, the process commences with target grounding, where a VLM anchors the user’s textual description to a specific visual instance in an initial video frame [33, 66]. To solve the association problem of maintaining the target’s identity across a distributed camera network, we employ a robust feature extractor (e.g., a Re-ID network) to match visual appearances [29]. Concurrently, a predictive model (e.g., a Kalman filter) estimates arrival times at downstream viewpoints, enabling just-in-time sensor activation to optimize resource usage [8, 77, 82]. Finally, the same VLM performs continuous verification, conducting frame-level reasoning to evaluate task predicates. This allows the system to intelligently decide when the query is satisfied, triggering early termination to conserve resources [4]. This unified loop reduces overhead by activating sensors only when needed and deactivating them once sufficient proof is obtained.

5 EVALUATION ON TOPOSENSE-BENCH

We conduct a comprehensive evaluation on our large-scale benchmark, *TopoSense-Bench*, to empirically validate the STG paradigm and quantify the performance of *IoT-Brain*. Our highlights are as follows:

- *IoT-Brain* delivers superior reliability and efficiency. On complex tasks, it boosts task success by up to 7.4× over the classical Hierarchical planner[24, 63] and runs nearly 2× faster with 6.6× fewer prompt tokens than the search-intensive Backtracking planner[14, 56] (§5.2).

- *IoT-Brain* scales and generalizes well. Its verification overhead grows near-linearly with query complexity, not exponentially. Furthermore, its performance gains persist across diverse foundation models, confirming the benefits stem from our architecture, not a specific backbone (§5.3).
- *IoT-Brain* derives strength from the full STG pipeline. Ablations confirm the synergy of its core components and reveal that unverified hypotheses are actively harmful, reinforcing “verify-before-commit” as a prerequisite for reliable physical-world planning (§5.4).

5.1 Experimental Setup

The TopoSense-Bench Benchmark. To rigorously evaluate our systems, we constructed *TopoSense-Bench*, a large-scale benchmark for the S^3 problem. Its central design principle is semantic textualization, which turns raw, machine-oriented OSM data into a structured knowledge base readable by LLMs. Concretely, we normalize and refine ontological tags[2], generate hierarchical human-readable names for all entities[1, 39], and project geodetic coordinates into a site-local Cartesian frame. The result is a unified semantic substrate where physical topology and sensor capabilities can be reasoned about jointly, closing the gap between cryptic geospatial encodings and natural language understanding.

The benchmark instantiates a realistic, topologically diverse university campus that serves as a large-scale testbed. It includes a comprehensive knowledge base spanning 33 buildings, 161 floor plans, and a dense network of 2,510 cameras. Layered on this environment is a carefully curated suite of 5,250 natural language queries that covers common user intents as well as challenging corner cases. Queries are organized into a three-tier taxonomy, detailed in Tab. 1, which supports fine-grained diagnosis of planning and reasoning capability across the full spectrum of S^3 challenges.

5.2 End-to-End Performance

Compared Paradigms. We benchmark *IoT-Brain* against agentized implementations of three influential LLM planning paradigms. To ensure comparability, all systems use the

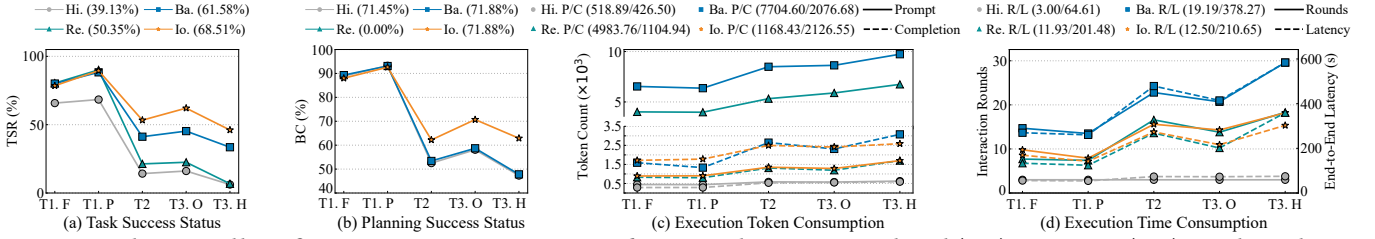


Figure 8: The overall performance on *TopoSense-Bench*. We evaluate Hierarchical (Hi.), Reactive (Re.), Backtracking (Ba.), and IoT-Brain (Io.) across five task categories, measuring task success rate, blueprint correctness, token usage, iteration rounds, and end-to-end latency. The legend displays the average performance for each paradigm.

Table 1: Statistics & Taxonomy of *TopoSense-Bench*.

Knowledge Base Statistics	Value
Buildings / Floor Plans / Outdoor Segments	33 / 161 / 53
Total Topological Scenarios / Deployed Cameras	7,832 / 2,510
Query Dataset Statistics	Count (%)
Tier 1: Intra-Zone Perception	
T1.a: Focal Scene Awareness e.g., "verify activity near the door to room 5F 1"	1,433 (27.3%)
T1.b: Panoramic Coverage e.g., "how many people are in the conference hall?"	1,129 (21.5%)
Tier 2: Intra-Building Coordination	
T2: Intra-Building Coordination e.g., "I lost a notebook between lab-8 and lab-7"	988 (18.8%)
Tier 3: Inter-Building Coordination	
T3.a: Open-Space Coordination e.g., "track my path from the west gate to the tennis court"	946 (18.0%)
T3.b: Hybrid Indoor-Outdoor e.g., "I walked from building-1 to building-2..."	754 (14.4%)

same foundation LLMs and API toolkits, differing only in their reasoning and execution policies. (1) The Hierarchical planner [24, 63] follows a decompose-then-execute strategy. It produces a complete, high-level plan upfront without intermediate verification, yielding speed at the cost of brittleness. (2) The Reactive planner [17, 80] instantiates the Thought-Action-Observation (TAO) loop. It operates step-by-step, making it highly adaptive but often myopic on long-horizon tasks. (3) The Backtracking planner [14, 56] is inspired by depth-first search over a decision tree. It improves reliability by exploring multiple branches, but this expanded search typically incurs prohibitive inference costs and can commit to locally optimal yet globally inefficient paths. Fig. 7 contrasts these workflows with our *STG* approach.

Evaluation Metrics. We assess each paradigm's performance from the dual perspectives of reliability and efficiency. We use a suite of standardized metrics: For reliability, we measure (1) task success rate (TSR), the percentage of queries yielding a functionally correct final answer, and (2) blueprint correctness (BC), which evaluates if a pre-execution plan is logically sound and topologically feasible, scored via an LLM-as-a-Judge protocol [84]. For efficiency, we quantify (3)

inference cost, measured in total LLM tokens; (4) interaction rounds, the number of agent-LLM turns; and (5) end-to-end latency, the total time from query submission to final answer. **Reliability Analysis.** Fig. 8(a-b) presents the TSR and BC across all task categories and difficulty tiers. A key observation is that while baseline planners often struggle to translate symbolically correct plans into real-world success, IoT-Brain consistently maintains high TSR, especially as task complexity increases to long-horizon settings. This empirically demonstrates the *STG* paradigm's effectiveness in resolving the fundamental gaps that plague end-to-end LLM planning.

A notable paradox emerges on simple Tier 1 tasks. The plan-less Reactive planner achieves a strong TSR, demonstrating for simple tasks, adaptive tool-use can suffice in easy settings. The Hierarchical planner, by contrast, highlights the critical gap between symbolic planning and physical execution. It frequently produces plausible blueprints with high BC scores, yet achieves markedly lower TSR. This discrepancy reflects the representation and reasoning gaps, where a plan is logically sound on paper proves physically unrealizable without grounding in the real world. This confirms a correct blueprint is necessary but not sufficient for success.

As complexity escalates to Tier 2 and 3, the necessity of structured grounding becomes indisputable. The performance of all baseline planners degrades sharply, which is a clear manifestation of the reasoning gap. Their inability to construct coherent, long-horizon plans leads to systemic failure. In stark contrast, on the most complex T3.Hybrid task, IoT-Brain achieves a 46.1% TSR. This not only represents a more than 7.4× improvement over the Hierarchical planner's 6.2% but also surpasses the strongest search-intensive alternative, the Backtracking planner, by 37.6%. This sustained performance is a direct result of *STG*'s "verify-before-commit" process, which constructs a globally coherent blueprint and systematically closes the reasoning gap.

Efficiency Analysis. Reliability must also be delivered efficiently. The computational overhead for each framework is reported in Fig. 8(c-d) show that the cost escalates with task complexity for unstructured planners, exposing an optimization gap. The Hierarchical planner is fast yet unreliable, while Reactive and Backtracking exhibit rapidly increasing

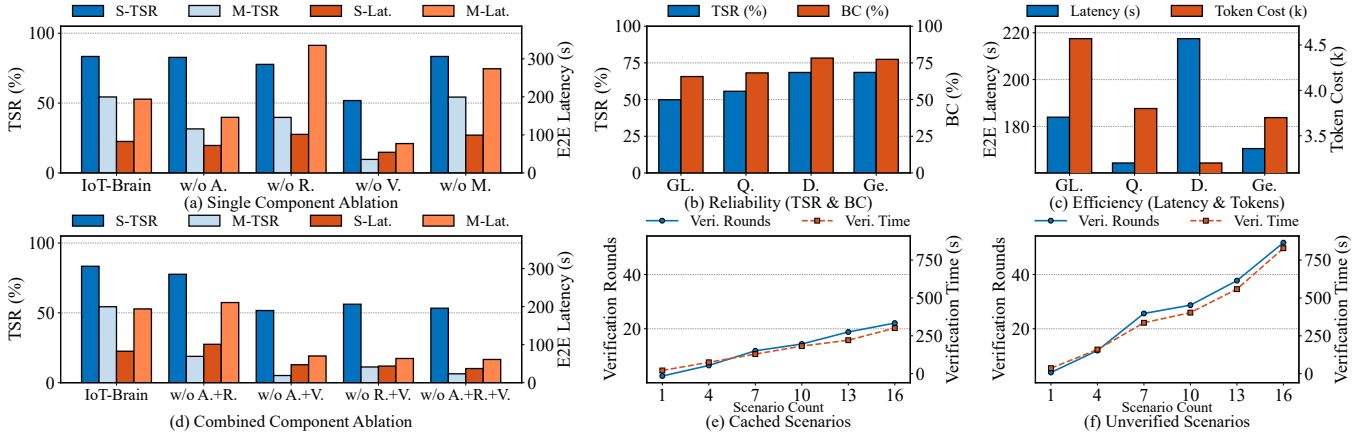


Figure 9: (a) & (d): Ablation study of *IoT-Brain*'s core components on single and multi scenario tasks. (A.: Anchor, R.: Reasoner, V.: Verifier, M.: Memory, S: Single-Scenario, M: Multi-Scenario). (b) & (c): Sensitivity to different LLMs. (GL.: GLM-4, Q.: Qwen-Max, D.: DeepSeek-V3, Ge.: Gemini-2.5-Flash). (e) & (f): Scalability with query complexity.

token consumption and latency as complexity grows. In pursuit of reliability through exhaustive search, Backtracking reaches nearly 600 s on the most complex Tier 3 task.

IoT-Brain closes this gap by casting planning as constrained optimization on a verified graph, yielding a markedly different efficiency profile. Its resource use grows more smoothly with complexity. The principled verification loop is more focused than the brute-force exploration of Backtracking and less myopic than the trial-and-error of Reactive. While delivering superior reliability, *IoT-Brain* is nearly 2× faster on the most complex tasks and uses, on average, 6.6× fewer prompt tokens than Backtracking. These results show that the *STG* paradigm not only improves reliability but also provides a scalable path to computational efficiency, achieving a win-win in both correctness and cost.

5.3 Scalability and Sensitivity Analysis

Sensitivity to Foundation Models. To demonstrate that the benefits of *STG* are not tied to a single proprietary model, we evaluate *IoT-Brain*'s performance across four foundation LLMs. As expected, more powerful models yield higher end-to-end reliability (Fig. 9(b)), confirming that the quality of the underlying LLM is a significant factor. More importantly, the efficiency profile shows no orders-of-magnitude variation in latency and token cost across models (Fig. 9(c)). While different APIs exhibit distinct latency characteristics, the token consumption remains relatively stable. These results indicate *STG* supplies a strong scaffold that channels reasoning into verifiable structure, allowing less capable models to perform competitively and confirming that observed gains arise from our architecture, not from any specific LLM's capabilities.

Scalability with Query Complexity. We investigate how *IoT-Brain*'s verification overhead scales with query complexity, a key determinant of real-world usability. For cached scenarios, verification overhead grows sublinearly, as the system

intelligently reuses entries from *Spatial Memory* (Fig. 9(e)). In stark contrast, for novel, unverified scenarios, the number of verification rounds also increases in a near-linear fashion with the count of distinct locations referenced in the query (Fig. 9(f)). This predictable and bounded growth stands in sharp contrast to the combinatorial blowup typical of unconstrained planning and demonstrates the inherent scalability of our structured, hypothesis-driven verification process.

5.4 Ablation Study

Impact of Individual Components. Fig. 9(a) shows the indispensable role of each component. Removing the *Grounding Verifier* is catastrophic, causing the multi-scenario TSR to plummet below 10%, confirming the necessity of a "verify-before-commit" loop for reliable planning. Removing the *Spatial Reasoner* also severely degrades performance, especially on complex tasks, as the *Verifier* is consequently forced into exhaustive checks without the *Reasoner*'s focused hypotheses. Similarly, omitting the *Topological Anchor* nearly halves multi-scenario TSR, underscoring its importance in structuring long-horizon plans. Finally, eliminating the *Memory* modules, while not impacting single-run TSR, nearly doubles latency in complex settings, demonstrating their critical value in efficiently amortizing verification costs.

Synergy of Combined Components. Fig. 9(d) underscores the components' tight complementarity. A variant lacking all core modules performs as poorly as the Hierarchical baseline, with its multi-scenario TSR collapsing to 6.4%, confirming that the full pipeline is indispensable for robust behavior. More tellingly, a variant that isolates the *Spatial Reasoner* by removing its structuring and verification counterparts performs even worse, with its multi-scenario TSR dropping to a mere 5.1%. This result reveals a crucial, counter-intuitive insight that the *Reasoner*'s unverified hypotheses are not

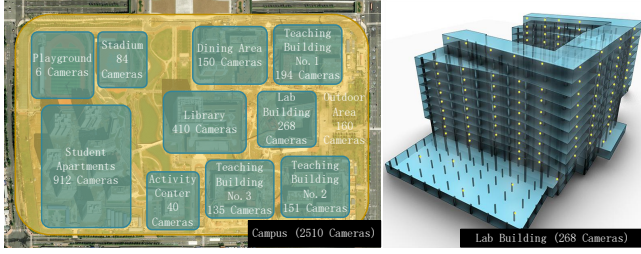


Figure 10: The real-world testbed environment.

merely neutral but actively harmful, introducing strong, misleading biases that derail the entire planning process. These results powerfully validate our philosophy that a structured, verifiable grounding process is not an add-on, but the fundamental prerequisite for reliable physical-world planning.

6 EVALUATION ON A PHYSICAL TESTBED

We complement our benchmark results with an end-to-end evaluation of *IoT-Brain* in a large-scale, physical testbed. The experiments confirm the *STG* paradigm’s effectiveness in a real-world deployment, yielding a near-optimal balance of reliability and resource efficiency. Our highlights are:

- *IoT-Brain* demonstrates strong resource efficiency, achieving a 49.84% TCR that approaches the reliability upper bound while using $4.1\times$ less bandwidth and processing $4.2\times$ fewer frames than the resource-agnostic method (§6.2).
- *IoT-Brain*’s architecture is both efficient and robust. Latency profiling confirms its planning engine is swift, with dominant costs arising from task-intrinsic complexity. Being model-agnostic, the framework allows practitioners to flexibly balance performance, cost, and privacy by integrating diverse VLM backbones (§6.3).

6.1 Testbed Configuration

Physical Environment. Our real-world testbed is a large-scale university campus instrumented with 2,510 Hikvision IP cameras distributed across 11 major areas. Coverage ranges from sparse road networks to dense indoor deployments, with the Lab Building alone hosting 268 cameras (see Fig. 10).

System Deployment. The *IoT-Brain* system runs on a centralized server equipped with two NVIDIA A100 GPUs. Its core planning engine utilizes the Gemini-2.5-Flash API[25], while the *Perception Aligner* executes on-premises using a compact and efficient visual stack that includes YOLOv8[68] for detection and PersonViT-B/16[31] for image re-identification. All components communicate over the campus Wi-Fi infrastructure, reflecting a practical deployment setting.

6.2 Real-World Deployment Results

Evaluation Protocol. We evaluated end-to-end performance on 587 annotated real-world trajectories across three paradigms. In addition to our *IoT-Brain* framework, we considered two

Table 2: Real-World Scheduling Paradigm Comparison.

Paradigm	TCR (%)	Latency (s)	Bandwidth (GB)	TFP (frames)
Static Scheduling	3.61	403.42	0.138	179
Naive Parallel	65.64	927.99	0.540	704
IoT-Brain (Ours)	49.84	413.69	0.131	166

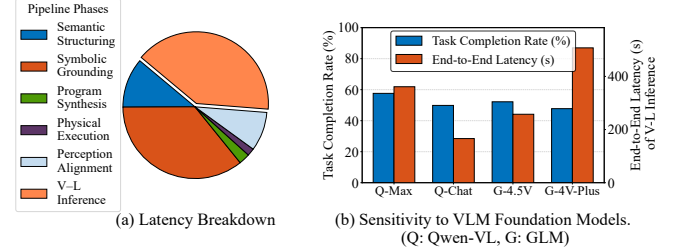


Figure 11: System performance analysis.

baselines to benchmark against both common practice and a theoretical optimum: (1) Static Scheduling, which mirrors conventional security practice by triggering downstream sensors using a constant-velocity pedestrian model[61], and (2) Naive Parallel Scheduling, a resource-agnostic upper bound that activates all potentially relevant cameras simultaneously. To ensure a controlled comparison, all vision–language queries were handled by a locally deployed Qwen-VL-Chat model[7]. Performance was measured using a comprehensive suite of metrics, including task completion rate (TCR), end-to-end latency, network bandwidth, and total frames processed (TFP). **Performance Analysis.** Tab. 2 highlights the trade-off between reliability and resource use. Static Scheduling is frugal yet brittle, reaching only 3.61% TCR and failing systematically whenever a target deviates from its pre-defined coverage. At the other extreme, Naive Parallel establishes an empirical upper bound on reliability at 65.64% TCR, but does so at untenable cost, consuming $4.1\times$ more bandwidth and processing $4.2\times$ more frames than our system.

IoT-Brain navigates this trade-off with a strong balance of reliability and thrift. Driven by the *STG* paradigm for intelligent planning and the *Perception Aligner* for just-in-time execution, it achieves a high TCR of 49.84% approaching the empirical upper bound, while keeping a resource footprint comparable to the far less reliable static strategy, using the least bandwidth and processing the fewest frames among all paradigms. Its sequential, plan-informed activation prevents the heavy data and processing burden of parallel operation. The results confirm that verifiable planning coupled with perception-aligned execution yields a near-optimal balance, delivering robust reliability with exceptional efficiency.

6.3 System Analysis and Sensitivity

Latency Breakdown. We profiled the end-to-end latency for the *IoT-Brain* pipeline to characterize its temporal behavior. The breakdown in Fig. 11(a) clarifies the cost structure of

complex semantic scheduling. The two dominant phases are *Symbolic Grounding* and *V-L Inference*, which account for most of the execution time on challenging Tier 2 & 3 tasks. The high cardinality of scenarios necessitates extensive verification cycles during grounding, while a larger scheduled sensor set increases the volume of frames for VLM inference. By contrast, the initial *Semantic Structuring* stage is remarkably efficient. Overall, this analysis reveals that the primary latency drivers are not inefficiencies in our planning engine, but rather the intrinsic complexity of the tasks, which demands substantial verification and perception effort.

Sensitivity to VLM Foundation Models. The framework's performance is influenced by the VLM selected within the *Perception Aligner*. To demonstrate architectural generality, we evaluate our framework with both local open-source and proprietary API-based VLMs. As shown in Fig. 11(b), the results reveal a clear performance-latency trade-off. API-based models such as Qwen-VL-Max achieve higher TCR but introduce significant latency, whereas the open-source Qwen-VL-Chat offers lower latency with a corresponding reduction in TCR. This outcome underscores our framework's robustness. The *STG* paradigm, by decoupling planning from perception, is fundamentally model-agnostic, providing a stable planning scaffold that functions effectively regardless of the underlying VLM. The choice of foundation model thus becomes a configurable trade-off for deployers, allowing them to balance performance, cost, and data privacy.

7 DISCUSSION

Generality and Limitations. Our current instantiation centers on a large-scale camera network, yet the *STG* paradigm is fundamentally sensor-agnostic. Its abstract graph representation allows nodes and edges to encode any spatially aware sensing modality, such as LiDAR[53] or microphone arrays[71], along with their topological, visibility, and geometric relationships. We emphasize cameras as they most clearly expose the formidable spatial reasoning demands of the S^3 problem. Nevertheless, two caveats remain. First, the system assumes a static world model. Deployment in dynamic settings with sensor outages or temporary obstacles will require continual world state updates[36, 37]. Second, our evaluation is confined to a single campus domain. Broader validation across diverse IoT scenarios (e.g., smart homes, city-scale traffic networks) and under distribution shifts remains a crucial direction for future work.

Privacy-by-Design. As pervasive sensing becomes widespread, protecting individual privacy is paramount [46, 48]. *IoT-Brain* adopts data minimization as a first principle. Intent-driven planning activates only the minimal set of relevant

sensors for the required interval, reducing exposure by design. The architecture also supports decoupled symbolic planning, where high-level verification operates in an abstract space without accessing raw sensor data [18, 62], while on-premises perception keeps sensitive content within a local trust domain [12]. Furthermore, the grounded graph enables transparent accounting, as every activation decision is tied to an explicit hypothesis and its verification record. These elements together provide a practical blueprint for privacy-preserving intelligent sensing systems aligned with organizational policy and regulatory requirements.

8 RELATED WORK

LLMs for Sensor System Control. Recent pioneering works explore using LLMs to translate human intent into sensor actions, typically in constrained smart-home settings [5, 35, 43]. These approaches often rely on formal grammars to enhance plan robustness. Our work differs fundamentally by addressing the challenges of scale and physical-world grounding, shifting focus from small-scale plan robustness to verifiable, resource-optimal scheduling in large-scale deployments.

LLM-based Agentic Planning. The advent of LLMs has catalyzed agentic planning frameworks like Hierarchical [24, 63], Reactive [65, 80], and Backtracking [9, 56, 79] paradigms. These approaches excel at reasoning within reliable digital substrates. Our work centers on grounding language-based reasoning in the noisy physical world. *STG* functions as a neurosymbolic scaffold constrains reasoning through verifiable graph construction, supplying the structural guarantees needed for reliability in high-stakes embodied settings.

Language-Guided Perception. Person re-identification traditionally focused on image-to-image retrieval [16, 85]. The introduction of language models enabled text-to-image re-identification and language-guided tracking in continuous video [41, 72]. Most studies presume a predefined stream or a broad search space. Our *Perception Aligner* instead operates within a proactively scheduled, on-demand feed, providing just-in-time verification to drive the next sensor activation rather than performing persistent, open-domain tracking.

9 CONCLUSION

We formalize *Semantic-Spatial Sensor Scheduling* (S^3) and reveal critical gaps in LLM planning. To address these, we introduce *STG*, a "verify-before-commit" neurosymbolic paradigm that decouples semantic inference from deterministic scheduling. Our implementation, *IoT-Brain*, evaluated on our *TopoSense-Bench* benchmark and real-world deployment, demonstrates significant gains in both reliability and efficiency over representative planners. These contributions provide a practical blueprint for LLMs to robustly translate high-level intent into correct physical action.

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